# Chapter 7. Nonlinear Regression

Lecture 6-4 (polynomials, step functions, basis functions, regression splines, smoothing splines, local regression, GAM)

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# 1 Basis-Function Regression

## 1.1 Polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i.$$

For large enough $d$ this produces an extremely nonlinear curve, yet the parameters are estimated by OLS: create $X_1 = X$, $X_2 = X^2$, ... and treat it as multiple linear regression. We are rarely interested in the coefficients; we care about the fitted function

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4,$$

with pointwise interval $\hat{f}(x_0) \pm t_{1-\alpha/2,\mathrm{df}}\,\mathrm{SE}[\hat{f}(x_0)]$, where $\mathrm{SE}[\hat{f}(x_0)]^2 = \ell_0^\top \widehat{\mathrm{Var}}(\hat{\beta})\ell_0$ with $\ell_0 = (1, x_0, \ldots, x_0^4)^\top$. Fix $d$ at a low value (rarely above 3 or 4) or choose it by CV. Example: a degree-4 polynomial of wage on age, with 95% bands; on the Auto data, $d = 1$ underfits mpg on horsepower, $d = 2$ fits well, $d = 14$ is wild.

**Limitation.** Polynomials are global: the fit in one region is affected by data far away, the tails become erratic for large $d$, and a polynomial may fail to recover a nonlinear pattern even when $d$ is huge.

## 1.2 Step functions

Cut the range of $X$ at $c_1 < \cdots < c_K$:

$$C_0(X) = \mathbb{I}(X < c_1),\ C_1(X) = \mathbb{I}(c_1 \le X < c_2),\ \ldots,\ C_K(X) = \mathbb{I}(c_K \le X),$$

so $C_0 + \cdots + C_K = 1$, and fit $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$ ($C_0$ is dropped because of the intercept). $\beta_0$ is the mean for $X < c_1$ and $\beta_k$ the difference of bin $k$ from it. With bins $< 35$, 35–50, 50–65, $\ge 65$ for wage on age, the first bin misses the increasing trend; still, step functions are popular in biostatistics and epidemiology (e.g. five-year age groups). The result depends on the breakpoints.

## 1.3 Basis functions

Both are special cases of

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

with fixed, known functions $b_k$: $b_j(x) = x^j$ for polynomials and $b_j(x) = \mathbb{I}(c_j \le x < c_{j+1})$ for step functions. Everything of linear regression (least squares, SEs, $F$-tests) applies. Other bases: Fourier series, wavelets (e.g. for image denoising), and the splines below.

# 2 Regression Splines

## 2.1 Piecewise polynomials

Fit separate low-degree polynomials over regions separated by **knots**; with one knot $c$,

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i, & x_i \ge c. \end{cases}$$

More knots give more flexibility. Unconstrained pieces jump at the knot, so we add constraints (continuity, continuity of derivatives). Splines have the "maximum" amount of continuity.

## 2.2 Linear splines

**Definition: linear spline**

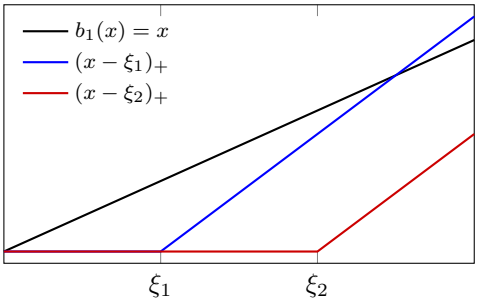
A linear spline with knots $\xi_1 < \cdots < \xi_K$ is piecewise linear and continuous at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \varepsilon_i, \qquad b_1(x) = x, \quad b_{k+1}(x) = (x - \xi_k)_+,$$

with $(x - \xi)_+ = x - \xi$ if $x > \xi$ and 0 otherwise.

Each truncated term $(x - \xi_k)_+$ adds a change of slope $\beta_{k+1}$ at $\xi_k$ while keeping continuity.

(a) basis functions




(b) linear spline

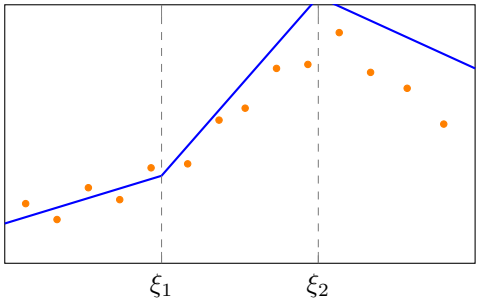



**Figure 1:** (a) The truncated power basis of a linear spline with knots $\xi_1, \xi_2$. (b) A fit $\beta_0 + \beta_1 x + \beta_2(x - \xi_1)_+ + \beta_3(x - \xi_2)_+$: continuous, with slope changes $\beta_2$ at $\xi_1$ and $\beta_3$ at $\xi_2$.

## 2.3 Cubic splines

**Definition: cubic spline and truncated power basis**

A cubic spline with knots $\xi_1, \ldots, \xi_K$ is piecewise cubic with continuous derivatives up to order 2 at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i,$$

$$b_1(x) = x, \; b_2(x) = x^2, \; b_3(x) = x^3, \; b_{k+3}(x) = (x - \xi_k)_+^3, \; k = 1, \ldots, K.$$

**Theorem: degrees of freedom of a cubic spline**

The space of cubic splines with $K$ knots has dimension $K + 4$, and the truncated power basis (with the constant) is a basis.

**Proof.** $K+1$ unconstrained cubic pieces have $4(K+1)$ parameters; continuity of the function, first and second derivatives at each of the $K$ knots imposes $3K$ linear constraints, leaving $4(K+1) - 3K = K + 4$. Each $(x - \xi_k)_+^3$ is twice continuously differentiable (its first two derivatives vanish at $\xi_k$) and its third derivative jumps by 6 at $\xi_k$, so the $K + 4$ functions are linearly independent splines. ■

With continuity alone the fit can have a kink; continuity of the first and second derivatives makes it visually smooth.

## 2.4 Natural cubic splines

Splines can have high variance at the outer range of the predictor (few data, wide confidence bands). A **natural cubic spline** additionally requires the function to be *linear* beyond the boundary knots.

This adds 4 constraints (second and third derivatives zero at both ends), so a natural cubic spline with $K$ knots has $K$ degrees of freedom (including the intercept), and its confidence bands are narrower at the boundaries.

### 2.5 Number and locations of knots

Decide the number $K$ (or the degrees of freedom) and place knots at quantiles of the observed $X$ (or let software do it), so that each region has a similar amount of data. Choose $K$ by cross-validation. On the Wage data, a degree-15 polynomial is wild at the boundaries, whereas a natural cubic spline with 15 df is well behaved: splines add flexibility through knots while keeping the degree fixed.

## 3 Smoothing Splines

**Definition: smoothing spline**

Among all twice-differentiable functions $g$,

$$\min_g \underbrace{\sum_{i=1}^{n}(y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness penalty}} .$$

The first term makes $g$ match the data; the second controls how wiggly $g$ is ($g''$ measures curvature), modulated by $\lambda \geq 0$. The smaller $\lambda$, the more wiggly the function, eventually interpolating the $y_i$ at $\lambda = 0$; as $\lambda \to \infty$, $g$ becomes linear (any linear function has zero penalty). $\lambda$ can be chosen by $K$-fold CV.

**Theorem: the minimizer is a natural cubic spline**

The minimizer $\hat{g}_\lambda$ is a natural cubic spline with knots at the unique values of $x_1, \ldots, x_n$ — a shrunken version of the interpolating natural spline. Consequently $\hat{g}_\lambda = (\hat{g}(x_1), \ldots, \hat{g}(x_n))^\top = S_\lambda y$ for an $n \times n$ smoother matrix $S_\lambda$.

**Idea.** Given any candidate $g$, let $\tilde{g}$ be the natural cubic spline interpolating the values $g(x_i)$. Then RSS is unchanged, and an integration-by-parts argument shows $\int \tilde{g}''^2 \leq \int g''^2$, with equality only if $g = \tilde{g}$. So the search can be restricted to natural cubic splines with knots at the $x_i$, which form an $n$-dimensional linear space $g(x) = \sum_j N_j(x)\theta_j$. The problem becomes $\min_\theta \|y - N\theta\|^2 + \lambda\theta^\top \Omega_N \theta$ with $(\Omega_N)_{jk} = \int N_j'' N_k''$, solved by $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$, a generalized ridge regression. Hence $S_\lambda = N(N^\top N + \lambda\Omega_N)^{-1} N^\top$. ■

The **effective degrees of freedom** $\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda)$ decreases from $n$ ($\lambda = 0$) to 2 ($\lambda \to \infty$) and is a convenient way to specify $\lambda$. Because $\hat{g}$ is a linear smoother, LOOCV costs one fit: $\mathrm{CV}(\lambda) = \frac{1}{n}\sum_i \{(y_i - \hat{g}_\lambda(x_i))/(1 - \{S_\lambda\}_{ii})\}^2$ (Ch. 3). On the Auto and Wage data, a smoothing spline with about 6.8 effective df (chosen by LOOCV) is nearly indistinguishable from a 16-df fit.

## 4 Local Regression

With a sliding weight function, fit separate (weighted) linear fits over the range of $X$.

**Algorithm: local regression at a target point $x_0$**

1. Gather the fraction $s = k/n$ of training points closest to $x_0$ (the **span**).
2. Assign weights $K_{i0} = K(x_i, x_0)$: zero outside the neighborhood, largest for the closest points (e.g. tricube weights).
3. Fit a weighted least-squares regression $\min_{\beta_0,\beta_1} \sum_i K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2$.
4. The fitted value is $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Choices: the weighting function, whether to fit a constant, linear or quadratic regression locally, and the span $s$, which plays the role of $\lambda$: small $s$ gives a wiggly fit, large $s$ a global one; choose $s$ by CV. Local linear fits remove the boundary bias that local constant (kernel) averages have. Like KNN, local regression suffers from the curse of dimensionality for more than three or four predictors.

## 5 Generalized Additive Models

GAMs allow flexible nonlinearities in several variables while retaining the additive structure of linear models:

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i.$$

A GAM can be fitted with natural splines (then it is one big least-squares regression on the combined basis), with smoothing splines or local regression, or mixing terms — some linear, some nonlinear, some step functions.

**Algorithm: backfitting (for smoothing-spline or local-regression terms)**

Initialize $\hat{\beta}_0 = \bar{y}$, $\hat{f}_j = 0$. Repeat until convergence: for $j = 1, \ldots, p$, smooth the partial residuals $r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik})$ against $x_{ij}$ to update $\hat{f}_j$, then center $\hat{f}_j$ to have mean zero.

Each $f_j$ is updated with the others held fixed, which is possible because the model is additive.

**Example: Wage data**

$\texttt{wage} = \beta_0 + f_1(\texttt{year}) + f_2(\texttt{age}) + f_3(\texttt{education}) + \varepsilon$, with natural splines for year and age and a separate constant for each education level. Holding the others fixed, wage rises slightly with year, rises and then falls with age, and increases with education.

**Pros.** A nonlinear $f_j$ for each $X_j$ without manually trying many transformations; the additive structure still allows examining the effect of each $X_j$ with the others fixed; potentially more accurate predictions. Effective degrees of freedom summarize each $f_j$. **Cons.** Additivity can miss important interactions (they can be added by hand, e.g. $f_{jk}(X_j, X_k)$ via a two-dimensional smoother).

## Summary

- Basis-function regression (polynomials, step functions, splines) stays within linear least squares.
- Splines are piecewise polynomials joined smoothly at knots: linear splines $(x - \xi)_+$, cubic splines $(x - \xi)_+^3$ with $K + 4$ df, natural cubic splines linear beyond the boundary ($K$ df).

- Smoothing splines $\min \mathrm{RSS} + \lambda \int g''^2$ yield natural cubic splines with knots at the data; $\mathrm{df}_\lambda = \mathrm{tr} S_\lambda$.
- Local regression fits weighted least squares in a sliding window (span $s$).
- GAMs combine univariate nonlinear functions additively (backfitting); interactions must be added explicitly.